MATH-1014 T10B

Tutorial 10: Infinite Sequences and Series

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Today's Roadmap

1. Concept Review: The Essence of Sequences and Series

2. Advanced Practice Problems

Review: The Fundamental Distinction

The most common point of failure is confusing a sequence with a series. We must rigorously distinguish between the terms themselves and their infinite accumulation.

Sequences (The Terms)

- An ordered list of numbers: $\{a_n\}_{n=1}^{\infty}$.
- **Convergence:** Evaluated directly via limits. If $\lim_{n\to\infty} a_n = L$, the sequence converges to L.
- **Key Tools:** Squeeze Theorem, L'Hôpital's Rule, and the Monotonic Sequence Theorem.

Series (The Accumulation)

- The infinite sum of a sequence: $\sum_{n=1}^{\infty} a_n$.
- **Convergence:** Evaluated by analyzing the sequence of *partial sums*, $S_n = \sum_{i=1}^{n} a_i$. If $\lim_{n\to\infty} S_n = s$, the series converges to s.

Review: The Test for Divergence (The Gatekeeper)

Before applying any complex convergence tests, always analyze the raw behavior of the n-th term.

The Test for Divergence

If $\lim_{n\to\infty} a_n$ does not exist, or if $\lim_{n\to\infty} a_n \neq 0$, then the series $\sum_{n=1}^{\infty} a_n$ is **divergent**.

The Ultimate Logic Trap

The converse of this theorem is **FALSE**.
If $\lim_{n\to\infty} a_n = 0$, it tells you **absolutely nothing** about the series. It might converge, or it might diverge.

- Example A: $a_n = \frac{1}{n} \to 0$, but $\sum \frac{1}{n}$ diverges (Harmonic Series).
- Example B: $a_n = \frac{1}{n(n+1)} \to 0$, and $\sum \frac{1}{n(n+1)}$ converges.

Review: The Integral Test & p-Series

When a_n behaves like a well-known continuous function $f(x)$, we can use integration to determine convergence.

The Integral Test Prerequisites

Let $a_n = f(n)$. You **must** verify three conditions on $[1, \infty)$ before integrating:

1. $f(x)$ is **continuous**.
2. $f(x)$ is **positive** ($f(x) > 0$).
3. $f(x)$ is **decreasing** ($f'(x) < 0$).

If verified, $\sum_{n=1}^{\infty} a_n$ and $\int_1^{\infty} f(x)dx$ share the same fate (both converge or both diverge).

Direct Consequence (p-Series): $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges if $p > 1$ and diverges if $p \leq 1$.
Rigor Note: $\sum a_n \neq \int f(x)dx$. The integral only tests convergence, it does not give the sum!

Problem 1: Squeeze Theorem for Sequences

Problem 1.

Determine if the following sequence converges or diverges. If it converges, strictly prove its limit.

$$a_n = \frac{1}{\sqrt{n^2+1}} + \frac{1}{\sqrt{n^2+2}} + \cdots + \frac{1}{\sqrt{n^2+n}}$$

Hint: This is a sum of n terms where both the number of terms and the terms themselves change with n. Direct limit evaluation fails. You must construct bounds and apply the **Squeeze Theorem**.

Solution 1: Bounding the Sequence

Step 1: Construct the Upper Bound (c_n)

To make the sum as large as possible, we replace every denominator with the *smallest* denominator in the sequence ($\sqrt{n^2+1}$).

$$a_n \leq \frac{1}{\sqrt{n^2+1}} + \frac{1}{\sqrt{n^2+1}} + \cdots + \frac{1}{\sqrt{n^2+1}}$$

Since there are exactly n terms:

$$c_n = \frac{n}{\sqrt{n^2+1}}$$

Step 2: Construct the Lower Bound (b_n)

To make the sum as small as possible, we replace every denominator with the *largest* denominator in the sequence ($\sqrt{n^2+n}$).

$$b_n = \frac{n}{\sqrt{n^2+n}}$$

So, we have established: $b_n \leq a_n \leq c_n$.

Solution 1: Limit Evaluation

Step 3: Evaluate limits of the bounds. For the lower bound:

$$\lim_{n\to\infty} b_n = \lim_{n\to\infty} \frac{n}{\sqrt{n^2+n}} = \lim_{n\to\infty} \frac{1}{\sqrt{1+\frac{1}{n}}} = 1$$

For the upper bound:

$$\lim_{n\to\infty} c_n = \lim_{n\to\infty} \frac{n}{\sqrt{n^2+1}} = \lim_{n\to\infty} \frac{1}{\sqrt{1+\frac{1}{n^2}}} = 1$$

Step 4: Conclusion via Squeeze Theorem.
Since $\lim_{n\to\infty} b_n = \lim_{n\to\infty} c_n = 1$, it strictly follows that:

$$\lim_{n\to\infty} a_n = \mathbf{1}$$

The sequence **converges**.

Problem 2: Recursive Monotonicity

Problem 2.
A sequence is defined recursively by $a_1 = \sqrt{6}$ and $a_{n+1} = \sqrt{6 + a_n}$.
Show that $\{a_n\}$ converges and find its limit.

Rigor Requirement

You cannot assume the limit exists and simply solve $L = \sqrt{6 + L}$. You must first **prove** it converges using the **Monotonic Sequence Theorem**.

Solution 2: Proving Convergence

Step 1: Prove it is bounded above by 3 (via Induction).

- Base case $(n = 1)$: $a_1 = \sqrt{6} < 3$. (True)
- Inductive step: Assume $a_k < 3$. Then $a_{k+1} = \sqrt{6 + a_k} < \sqrt{6 + 3} = \sqrt{9} = 3$.
- Therefore, $a_n < 3$ for all $n \geq 1$.

Step 2: Prove it is strictly increasing. We analyze $a_{n+1}^2 - a_n^2$:

$$a_{n+1}^2 - a_n^2 = (6 + a_n) - a_n^2 = -(a_n^2 - a_n - 6) = -(a_n - 3)(a_n + 2)$$

Since a_n is positive and strictly bounded above by 3 (from Step 1), $(a_n - 3) < 0$ and $(a_n + 2) > 0$. Thus, $-(a_n - 3)(a_n + 2) > 0$. This means $a_{n+1}^2 > a_n^2$, and since terms are positive, $a_{n+1} > a_n$. The sequence is **increasing**.

Solution 2: Finding the Limit

Step 3: Apply Theorem and Solve.
Because $\{a_n\}$ is both monotonic (increasing) and bounded (above by 3), the **Monotonic Sequence Theorem** guarantees that the limit $L = \lim_{n\to\infty} a_n$ exists.
Now we can safely take the limit of the recurrence relation:

$$\lim_{n\to\infty} a_{n+1} = \lim_{n\to\infty} \sqrt{6 + a_n}$$

$$L = \sqrt{6 + L}$$

Square both sides:

$$L^2 = 6 + L \implies L^2 - L - 6 = 0 \implies (L-3)(L+2) = 0$$

Since $a_n > 0$ for all n, the limit must be positive. Therefore:

$$\lim_{n\to\infty} a_n = \mathbf{3}$$

Problem 3: Advanced Telescoping Series

Problem 3.
Determine whether the following series converges or diverges. If it converges, find its exact sum.

$$\sum_{n=1}^{\infty} \arctan\left(\frac{1}{n^2+n+1}\right)$$

Hint: This is a hidden telescoping sum. Recall the trigonometric identity for the tangent of a difference: $\tan(\alpha-\beta) = \frac{\tan\alpha - \tan\beta}{1+\tan\alpha\tan\beta}$.

Solution 3: Algebraic/Trigonometric Manipulation

Step 1: Rewrite the argument to match the tangent difference identity.
Notice that the denominator can be factored strategically:

$$\frac{1}{n^2+n+1} = \frac{1}{1+n(n+1)} = \frac{(n+1)-n}{1+(n+1)(n)}$$

Step 2: Apply the arctangent identity.
Let $\alpha = \arctan(n+1)$ and $\beta = \arctan(n)$. Then $\tan\alpha = n+1$ and $\tan\beta = n$.

$$\tan(\alpha-\beta) = \frac{(n+1)-n}{1+(n+1)n}$$

Taking the arctangent of both sides yields:

$$\arctan\left(\frac{1}{n^2+n+1}\right) = \arctan(n+1) - \arctan(n)$$

This is now a perfect telescoping form!

Solution 3: The Partial Sums

Step 3: Expand the N-th partial sum S_N.

$$S_N = \sum_{n=1}^{N} [\arctan(n+1) - \arctan(n)]$$

$$\begin{aligned} S_N = &(\arctan(2) - \arctan(1)) \\ &+(\arctan(3) - \arctan(2)) \\ &+ \dots \\ &+(\arctan(N+1) - \arctan(N)) \end{aligned}$$

Massive cancellation occurs, leaving only the last part of the N-th term and the first part of the 1st term:

$$S_N = \arctan(N+1) - \arctan(1)$$

Step 4: Take the limit as $N \to \infty$.

$$\lim_{N\to\infty} S_N = \lim_{N\to\infty} (\arctan(N+1) - \frac{\pi}{4}) = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}$$

Problem 4: Rigorous Integral Test

Problem 4.
Use the Integral Test to determine if the following series converges or diverges:

$$\sum_{n=2}^{\infty} \frac{e^{1/n}}{n^2}$$

Warning

Before evaluating the improper integral, you **must** rigorously verify the three prerequisites of the Integral Test on the interval $[2, \infty)$ using Calculus.

Solution 4: Verifying Prerequisites

Step 1: Check Integral Test conditions for $f(x) = \frac{e^{1/x}}{x^2}$ on $[2, \infty)$.

- **Continuous:** Yes, the only discontinuity is at $x = 0$, which is outside $[2, \infty)$.
- **Positive:** Yes, $x^2 > 0$ and $e^{1/x} > 0$ for all $x \geq 2$.
- **Decreasing:** We must check the first derivative. Use the product rule on $f(x) = x^{-2}e^{x^{-1}}$:

$$f'(x) = -2x^{-3}e^{x^{-1}} + x^{-2}e^{x^{-1}}(-x^{-2}) = -x^{-3}e^{1/x}\left(2 + \frac{1}{x}\right)$$

  Since $x \geq 2$, every component here is positive, but the leading negative sign makes $f'(x) < 0$. Thus, $f(x)$ is strictly **decreasing**.

All prerequisites are met. We can proceed with the integral.

Solution 4: Evaluating the Integral

Step 2: Evaluate the Improper Integral.

$$I = \int_2^\infty \frac{e^{1/x}}{x^2} dx = \lim_{t\to\infty} \int_2^t x^{-2} e^{1/x} dx$$

Let $u = \frac{1}{x}$, then $du = -x^{-2}dx$, which means $-du = x^{-2}dx$. Change the limits of integration: When $x = 2$, $u = 1/2$. When $x = t$, $u = 1/t$.

$$I = \lim_{t\to\infty} \int_{1/2}^{1/t} -e^u du = \lim_{t\to\infty} [-e^u]_{1/2}^{1/t}$$

$$I = \lim_{t\to\infty} \left(-e^{1/t} - (-e^{1/2})\right)$$

As $t \to \infty$, $1/t \to 0$, so $e^{1/t} \to e^0 = 1$.

$$I = -1 + e^{1/2} = \mathbf{\sqrt{e} - 1}$$

Conclusion: Because the improper integral converges to a finite number, by the **Integral Test**, the original series $\sum_{n=2}^\infty \frac{e^{1/n}}{n^2}$ also **converges**.

Problem 5: The Integral-Sequence-Series Triad

Problem 5.
Let a sequence $\{a_n\}$ be defined by the definite integral:

$$a_n = \int_n^{n+1} xe^{-x}\,dx$$

Find the exact sum of the infinite series $\sum_{n=1}^{\infty} a_n$.

Hint: This problem synthesizes Integration by Parts, sequences, and series. Do not attempt to integrate the infinite sum directly. First, evaluate the integral to find an explicit formula for a_n, and carefully observe its structure.

Solution 5: Evaluating the Integral

Step 1: Integration by Parts.
We first find the antiderivative of $f(x) = xe^{-x}$. Let $u = x$ and $dv = e^{-x}dx$. Then $du = dx$ and $v = -e^{-x}$.

$$\begin{aligned} \int xe^{-x}dx &= -xe^{-x} - \int -e^{-x}dx \\ &= -xe^{-x} - e^{-x} \\ &= -(x+1)e^{-x} \end{aligned}$$

Step 2: Apply the limits of integration.

$$\begin{aligned} a_n &= \left[-(x+1)e^{-x}\right]_n^{n+1} \\ &= -(n+2)e^{-(n+1)} - \left(-(n+1)e^{-n}\right) \\ &= \mathbf{(n+1)e^{-n} - (n+2)e^{-(n+1)}} \end{aligned}$$

Solution 5: The Telescoping Series

Step 3: Analyze the series structure.
Let $b_n = (n+1)e^{-n}$. We can see that $a_n = b_n - b_{n+1}$.
This means the series $\sum_{n=1}^{\infty} a_n$ is a **Telescoping Series**.
Step 4: Compute the N-th partial sum S_N.

$$
\begin{aligned}
S_N &= \sum_{n=1}^{N} (b_n - b_{n+1}) \\
&= (b_1 - b_2) + (b_2 - b_3) + \cdots + (b_N - b_{N+1}) \\
&= b_1 - b_{N+1} \\
&= 2e^{-1} - (N+2)e^{-(N+1)}
\end{aligned}
$$

Solution 5: Taking the Limit

Step 5: Evaluate the limit as $N \to \infty$. To find the sum of the series, we take the limit of S_N. The second term requires L'Hôpital's Rule because it is an $\frac{\infty}{\infty}$ indeterminate form:

$$\begin{aligned}\lim_{N\to\infty}(N+2)e^{-(N+1)} &= \lim_{N\to\infty}\frac{N+2}{e^{N+1}} \\ &\overset{L'H}{=} \lim_{N\to\infty}\frac{1}{e^{N+1}} \\ &= 0\end{aligned}$$

Thus, the infinite series converges to:

$$\sum_{n=1}^{\infty} a_n = \lim_{N\to\infty} S_N = 2e^{-1} - 0 = \frac{\mathbf{2}}{\mathbf{e}}$$

Problem 6: The Sequence limit & Riemann Sum

Problem 6.
Evaluate the limit of the following sequence:

$$a_n = \frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{2n}$$

Strategic Shift

This is not a geometric series, and applying the Squeeze Theorem here yields bounds that are too loose to be useful. Recall from the early stages of integration (and the slides) that an infinite sum limit forms the very definition of a **definite integral**.

Solution 6: Revealing the Riemann Sum

Step 1: Algebraic Factorization.
We want to express a_n in the standard Riemann Sum format: $\sum f(x_i)\Delta x$. First, factor out $\frac{1}{n}$ from the entire sequence:

$$\begin{aligned} a_n &= \frac{1}{n}\left(\frac{n}{n+1} + \frac{n}{n+2} + \cdots + \frac{n}{n+n}\right) \\ &= \frac{1}{n}\left(\frac{1}{1+1/n} + \frac{1}{1+2/n} + \cdots + \frac{1}{1+n/n}\right) \end{aligned}$$

Step 2: Sigma Notation.
We can now compactly write this sequence as:

$$a_n = \sum_{i=1}^{n} \frac{1}{1+i/n} \cdot \left(\frac{1}{n}\right)$$

Solution 6: Transition to Calculus

Step 3: Map to the Definite Integral.
By analyzing the sigma notation, we can identify the components of a Right-Endpoint Riemann Sum on the interval $[0, 1]$:

- The width of each subinterval is $\Delta x = \frac{1-0}{n} = \frac{1}{n}$.
- The sample points are $x_i = 0 + i\Delta x = \frac{i}{n}$.
- The function being evaluated is $f(x) = \frac{1}{1+x}$.

Step 4: Evaluate the Integral.
Taking the limit as $n \to \infty$ converts the sum into a definite integral:

$$\begin{aligned} \lim_{n\to\infty} a_n &= \int_0^1 \frac{1}{1+x} dx \\ &= [\ln|1+x|]_0^1 \\ &= \ln(2) - \ln(1) = \ln \mathbf{2} \end{aligned}$$

THANK YOU!